

A Comparison of Back Grinding Processes for Bare Silicon and Through-Silicon Via Wafers Using Numerical Simulations

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Abstract—The optimization of grinding parameters for silicon wafers is necessary in order to reduce the likelihood of residual stresses and crack nucleation in the machined surface and improve the reliability of electronic packages. This paper describes numerical simulations performed to characterize the back grinding process for bare silicon wafers and through-silicon via (TSV) wafers using the finite element code ABAQUS. The grinding of the two wafer types was performed by simulating the motion of a diamond particle cutting through successive silicon and TSV layers. The silicon material was modeled using orthotropic elasticity and isotropic plasticity, while the copper vias were modeled using isotropic elasticity and Johnson-Cook plasticity. The polyethylene terephthalate (PET) material used as a backing tape for the silicon wafer was modeled using the Mooney-Rivlin hyperelastic model. The computed residual stresses and the plastic deformation in the superficial layer of the ground wafer were compared with experimental values and good correlation was observed. The simulation results can be used to better understand the local stresses and strain fields in both bare silicon and TSV wafers, and to quantify the effect of the copper vias on the wafer properties during and after grinding.

Keywords—Copper; finite element; grinding; modeling; silicon; TSVs; wafer

INTRODUCTION

The development of electronic devices is based on strict weight and size requirements. Smaller, lighter, and higher capacity devices at low cost have become expected end-user demands. In order to achieve this goal, electronic packaging plays a major role in the electronics industry. The thickness of silicon wafers directly affects the package size, so that the thinner the wafer, the smaller the electronic package. The goal of this study was to simulate the grinding operation of silicon wafers in order to achieve a better understanding of the parameters that influence the local stress and strain fields induced during this operation.

Copper TSVs can be inserted in silicon wafers as a means of minimizing the power consumption and increasing the speed of data transfer through the package. The manufacture of TSV wafers faces many challenges, one of them being the thinning

operation. This process is performed during the back grinding of the wafer after the copper vias are inserted into the wafer. Due to the difference in stiffness and hardness between silicon and copper, the grinding operation of thin wafers poses significant manufacturing challenges. The grinding process of hard silicon wafers with inserted soft copper vias oftentimes leads to via fracture and copper adhesion to the grinding wheel, a phenomenon also known as wheel loading. Several research studies have been devoted to the development and improvement of this process. Most of the studies performed were experimental or analytical in nature, with few analyses considering numerical simulations to investigate the wafer-wheel interaction [1-4]. In order to better understand the details of the wafer grinding process, a micron scale study is needed to clarify the internal stresses, strains, and deformations that occur in the wafer material [5, 6].

In this paper, a numerical study was performed to simulate the grinding of bare (no TSVs) silicon wafers, and the results were compared with experimental data. Based on the bare silicon grinding model, a TSV wafer model was developed. The results from the two models were compared with each other to better understand the effect of the copper via on the residual stresses induced during grinding. The two models showed good correlation with experimental data, and revealed differences in the residual stresses due to the presence of the copper via. These simulations can represent a foundation for further parametric studies to optimize the grinding process and improve the quality of ground silicon wafers.

FINITE ELEMENT MODEL

The goal of this study is to simulate and compare the back grinding models for bare silicon and TSV wafers. By comparing the two models, the effect of the copper vias on the residual stresses and the plastic deformation in the wafer can be better understood and quantified. Two finite element models were built in ABAQUS. The first model simulates the back grinding of bare silicon wafers. The second model simulates the back grinding process for TSV wafers. Two types of finite element grinding models have so far been studied in the literature. These models can be categorized by the scale at which the simulation is performed as macroscale and microscale models. The macroscale models consider the overall wheel-workpiece interaction, which captures the aggregate effects of

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the abrasive wheel on the machined surface [7]. The microscale models focus on the individual grain-material interactions, which can elucidate in greater detail the actual material removal mechanism. Microscale models are better suited to estimate the grinding forces directly without resorting to measurements or phenomenological modeling. The present model simulates the microscale grinding process, which includes the effect of a single diamond crystal (abrasive grain) while it removes layers from the silicon wafer as shown in Fig. 1.

A. Bare Silicon Wafer Model

A 2D model was built to simulate a portion of the silicon wafer and a small particle of diamond which moves horizontally and cuts through successive silicon layers. The model considers only a part of the silicon wafer which is completely attached to a plastic PET backing film as shown in Fig. 2. In the present simulations, the number of the cutting passes chosen was 12. The cutting depth of each pass was set at $2\text{ }\mu\text{m}$, which led to a realistic cutting depth for the total grinding time simulated. After the completion of 12 cutting passes, the model was allowed a relaxation period.

The industrial process of back grinding of silicon wafers is often performed in two steps. The first step is the coarse grinding process, where the bulk of the silicon material removal occurs. The second step is the fine grinding process, which uses a finer abrasive wheel to remove the damage from the coarse grind and where the wafer thickness is reduced to the desired value. In this work, the second step is simulated. The boundary conditions applied to the geometrical model of the wafer are shown in Fig. 2. The bottom of the model is pinned to simulate the attachment of the wafer to the vacuum chuck. The sides of the model are fixed using symmetry boundary conditions along the x axis to simulate the effect of the remaining parts of the wafer and the adhesive tape on each side, which have not been included in the geometrical model. The diamond crystal was displaced in the x direction to simu-

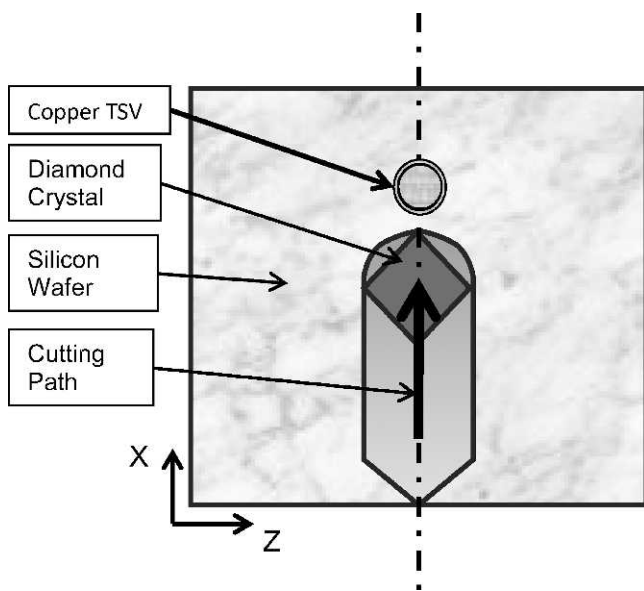


Fig. 1. Microscale model for the grinding of silicon wafers with inserted copper TSV.

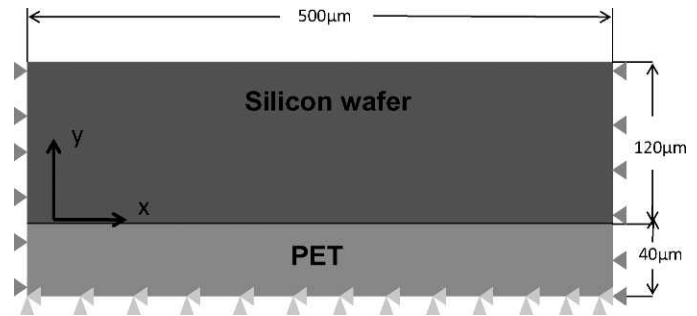


Fig. 2. Finite element model used to simulate the microscale grinding for bare silicon wafers mounted on a PET backing tape.

late the cutting action, and was displaced in the y direction to simulate the cutting depth.

An initial mesh with a global element size of $5\text{ }\mu\text{m}$ was used for both silicon and PET material. The mesh was further refined to a global element size of $1\text{ }\mu\text{m}$. Finally, a more refined mesh of $0.5\text{ }\mu\text{m}$ global element size was used for the silicon wafer, and a mesh with a larger element size of $20\text{ }\mu\text{m}$ was used for the PET adhesive tape. This mesh size was used to extract the data presented in this study.

B. TSV Wafer Model

A 2D model was built to simulate a part of the silicon wafer and a small diamond particle which removes silicon from a wafer with one copper via inserted in the middle of the silicon model. The wafer width is $500\text{ }\mu\text{m}$, and the initial silicon and copper via depth is $120\text{ }\mu\text{m}$. The diameter of the copper via is $10\text{ }\mu\text{m}$. The TSV is surrounded by a tantalum coating with a thickness of $0.1\text{ }\mu\text{m}$ and a silicon dioxide coating with a thickness of $0.5\text{ }\mu\text{m}$. The wafer is completely attached to a plastic PET backing film that is $150\text{ }\mu\text{m}$ thick, as shown in Fig. 3.

The number of cutting passes chosen was nine for this model. The cutting depth of each pass was set at $0.5\text{ }\mu\text{m}$, which represents a realistic cutting depth for a typical grinding process. After the completion of nine cutting passes, the model was allowed a relaxation period represented by the elastic unloading. For the bottom and sides of the wafer, similar boundary conditions to the bare silicon wafer model were applied.

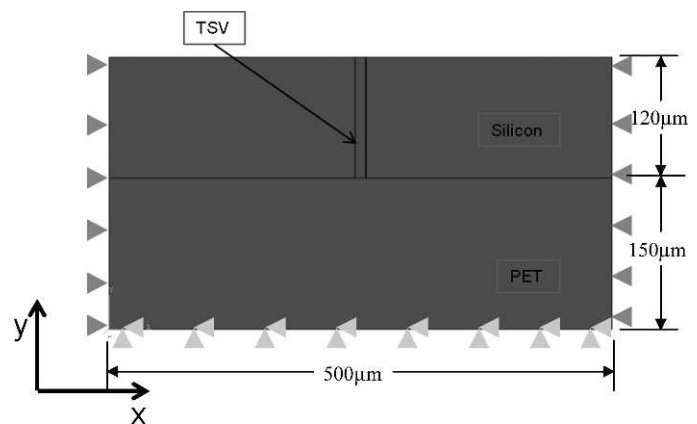


Fig. 3. Finite element model for the microscale grinding of TSV wafers.

C. Material Models

Because of the crystalline nature of the silicon single crystal, an orthotropic model was used to represent the elastic behavior of the material. The modeled wafer represented a silicon single crystal with an orientation of $\langle 100 \rangle$ in the x direction. The elastic constants can be a function of temperature and other predefined parameters, such as the strain rate. In this model, no temperature effect was considered, due to the cooling process that keeps the wafer at approximately 23°C. It is expected that the temperature may increase locally at the abrasive tool tip compared with the rest of the wafer; however, it is difficult to measure the local temperature. Moreover, any local spike in temperature occurs quickly, as the abrasive tip of the diamond wheel moves quickly over different areas of the wafer. In this study, the assumption that the entire wafer is kept at room temperature was made due to the action of the cooling liquid. Future studies should focus on the effect of the local temperature on the properties of the ground wafer. Also, no strain rate effect on the elastic behavior was considered in this study. The stress-strain relation takes the form

$$\begin{pmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{13} \\ \sigma_{23} \end{pmatrix} = \begin{pmatrix} D_{1111} & D_{1122} & D_{1133} & 0 & 0 & 0 \\ D_{1122} & D_{2222} & D_{2233} & 0 & 0 & 0 \\ D_{1133} & D_{2233} & D_{3333} & 0 & 0 & 0 \\ 0 & 0 & 0 & D_{1212} & 0 & 0 \\ 0 & 0 & 0 & 0 & D_{1313} & 0 \\ 0 & 0 & 0 & 0 & 0 & D_{2323} \end{pmatrix} \begin{pmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \epsilon_{33} \\ \gamma_{12} \\ \gamma_{13} \\ \gamma_{23} \end{pmatrix} \quad (1)$$

where the elastic constants D are presented in Table I [8].

The plastic behavior of the silicon wafer was simulated using an isotropic plasticity model with a von Mises hardening criterion. At room temperature, silicon does not deform significantly in the plastic region before damage onset and fracture. The plasticity of the silicon was modeled by building the effective stress-strain curve for a bilinear isotropic hardening curve with the points on the effective - diagram given in Table II.

The grinding does result in the damage and removal of silicon material. A shear damage initiation criterion with an element deletion scheme was used to simulate material removal during grinding. The shear damage criterion postulates the onset of damage due to shear band localization, and is used in conjunction with the von Mises plasticity model and isotropic hardening [9].

Table I
 D^{el} Matrix Constants

D^{el}	Matrix constant
D_{1111}	165,800 MPa
D_{1122}	63,740 MPa
D_{2222}	165,800 MPa
D_{1133}	63,740 MPa
D_{2233}	63,740 MPa
D_{3333}	79,620 MPa
D_{1212}	63,740 MPa
D_{1313}	63,740 MPa
D_{2323}	63,740 MPa

Table II
Equivalent Plastic Stress-Strain Curve for Silicon

Stress (MPa)	Plastic strain
1,000	0
1,100	0.005
1,200	0.007

The shear damage criterion model assumes that the equivalent plastic strain at the onset of damage $\bar{\epsilon}_s^{pl}$ is a function of the shear stress ratio θ_s , and strain rate $\dot{\bar{\epsilon}}^{pl}$

$$\theta_s = \frac{(q + k_s p)}{\tau_{\max}} \quad (2)$$

where $\tau_{\max}$ is the maximum shear stress, k_s is a material parameter, which was set to 0.3 according to [10], q is the equivalent stress, and p is the hydrostatic pressure. The criterion for damage initiation is met when the following condition is satisfied

$$w_s = \int \frac{d\bar{\epsilon}^{pl}}{\bar{\epsilon}_s^{pl}(\theta_s, \dot{\bar{\epsilon}}^{pl})} = 1 \quad (3)$$

In the above equation, w_s is a state variable calculated in the FEA code ABAQUS as a function of the plastic deformation, and this variable increases with the accumulation of plastic deformation. At each time step during the analysis, the increment in damage is calculated according to eq. (4). When w_s reaches a critical damage value, the element is deleted from the mesh [11].

$$\Delta w_s = \frac{\Delta \bar{\epsilon}^{pl}}{\bar{\epsilon}_s^{pl}(\theta_s, \dot{\bar{\epsilon}}^{pl})} \geq 0 \quad (4)$$

The copper via was modeled as an isotropic elastic material with a modulus of elasticity of 70 GPa and Poisson's ratio of 0.34. The empirically-based Johnson-Cook (J-C) model was used in these simulations to describe the plasticity of copper at various strain rates and temperatures

$$\bar{\sigma} = [A + B(\bar{\epsilon}^{pl})^n] \left[1 + c \ln \left(\frac{\dot{\bar{\epsilon}}^{pl}}{\dot{\epsilon}_0} \right) \right] (1 - \hat{\theta}^m) \quad (5)$$

where A , B , n , C , and m are material-dependent constants given in Table III, $\dot{\bar{\epsilon}}^{pl}$ is the effective plastic strain rate, and $\dot{\epsilon}_0$ is the normalized reference strain rate [12]. In this grinding process, the cooling system maintains the copper TSVs at near room temperature, thus the variation in temperature was not taken into account.

PET is a hyperelasticpolymeric material used as a thin layer to model the tape on which the wafer is mounted, before both

Table III
Johnson-Cook Constants for Copper

A (MPa)	90
B (MPa)	292
n	0.31
C	0.025
m	1.09

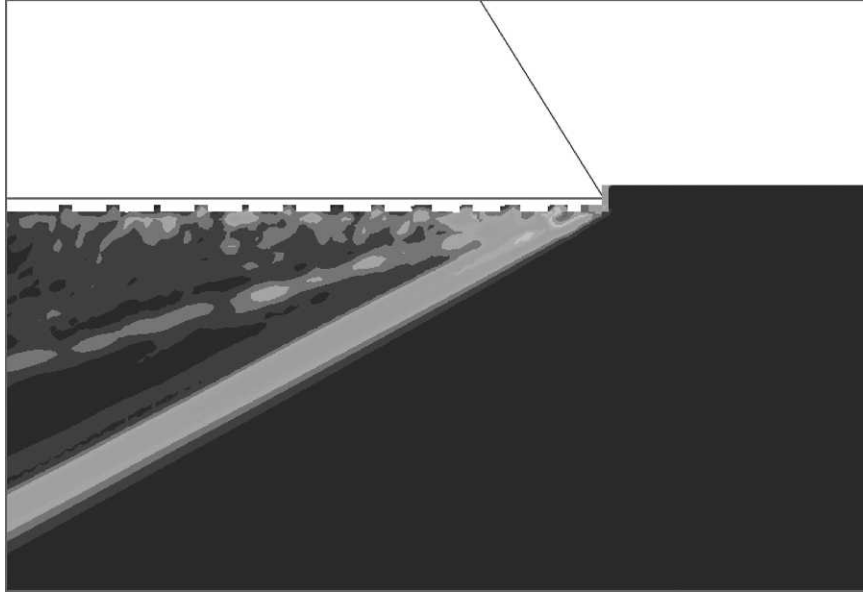


Fig. 4. Shear band localization angle ϕ measured between the surface of the workpiece and the shear band was 30° with a rake angle of 35° for the cutting diamond crystal.

wafer and tape are mounted on the vacuum chuck as one assembly. In order to simulate the behavior of this material, the Mooney-Rivlin model was used

$$U = C_{10}(\bar{I}_1 - 3) + C_{01}(\bar{I}_2 - 3) + \frac{1}{D_1}(J^{el} - 1)^2 \quad (6)$$

where U is the strain energy per unit of reference volume, C_{10} , C_{01} , and D_1 are temperature-dependent material parameters, and $\bar{I}_1$ and $\bar{I}_2$ are the first and second deviatoric stretch invariants defined as

$$\bar{I}_1 = \bar{\lambda}_1^2 + \bar{\lambda}_2^2 + \bar{\lambda}_3^2 \quad \text{and} \quad \bar{I}_2 = \bar{\lambda}_1^{-2} + \bar{\lambda}_2^{-2} + \bar{\lambda}_3^{-2} \quad (7)$$

where $\bar{\lambda}_1$, $\bar{\lambda}_2$, and $\bar{\lambda}_3$ are the three principal stretches, and J^{el} is the elastic volume ratio, which relates the total volume ratio to

the thermal volume ratio. The constant D_1 was set to a large number to eliminate the effect of the thermal expansion because of the effect of the cooling process. The constants C_{10} and C_{01} were chosen as 0.3. These values were found from the literature by curve fitting experimental data with the Mooney-Rivlin strain energy potential model used by ABAQUS [13].

The elasticity of the silicon dioxide was simulated using the isotropic elasticity model with a modulus of elasticity of 60 GPa, Poisson's ratio of 0.18, and a yield stress of 8400 MPa. Silicon dioxide is a hard material with little plasticity that was neglected in the present simulations.

Likewise, tantalum was modeled as an elastic material with a modulus of 186 GPa, Poisson's ratio of 0.34, and a yield stress of 170 MPa. The plasticity was simulated using an isotropic plasticity model.

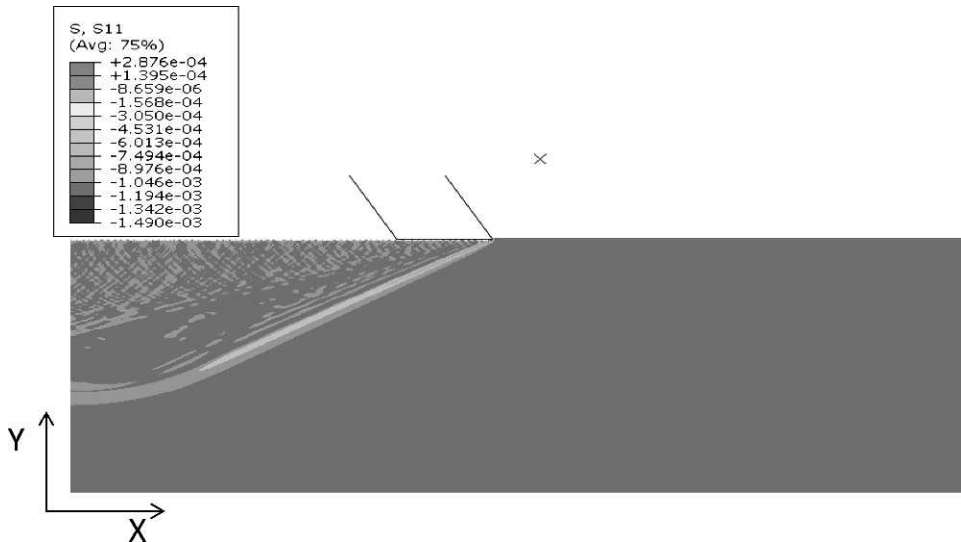


Fig. 5. Plot of the stress contours in the x direction during the cutting operation indicating the shear band formation. Stress units are in $\text{N}/\mu\text{m}^2$.

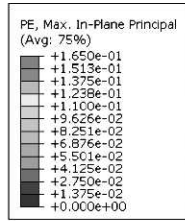


Fig. 6. Contour plot of the plastic strains in the wafer after grinding.

RESULTS

A. Bare Silicon Wafer Model

During grinding, shear band localization was observed to form at an angle of 30° with the surface of the specimen. A similar shear angle (ϕ) was reported in the literature in reference to the machining of metals. Comparing the values of ϕ computed in this model and the data from the literature [14], a similar behavior to that of other brittle materials can be found, as shown in Figs. 4 and 5. Fig. 4 also shows the surface elements that experienced damage after grinding. The damage is due to intense stresses induced by the grinding operation on the surface of the specimen. In practice, the damage results in surface cracks and the usual surface roughness. These surface cracks and roughness are later removed during fine grinding, which is not simulated here. Fig. 5 illustrates the stress distribution in the x direction and the shear stress band formation in the model.

The extent of plastic deformation at the backside of the wafer showed a good correlation when compared with literature data for ground wafers, as illustrated in Fig. 6. This figure also shows that the depth of the amorphous layer was found to be less than $5 \mu\text{m}$, which corresponds to a value reported in the

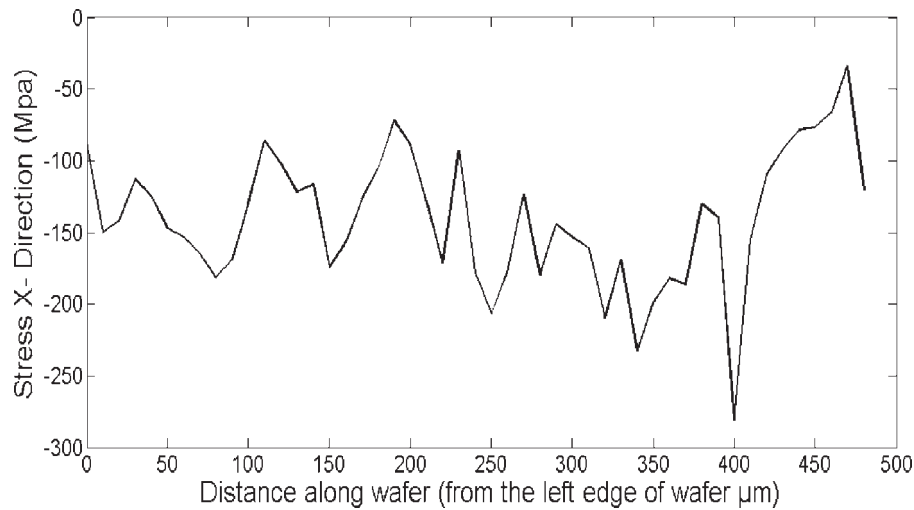


Fig. 7. Simulated stress σ_x variations at the back of the wafer after grinding.

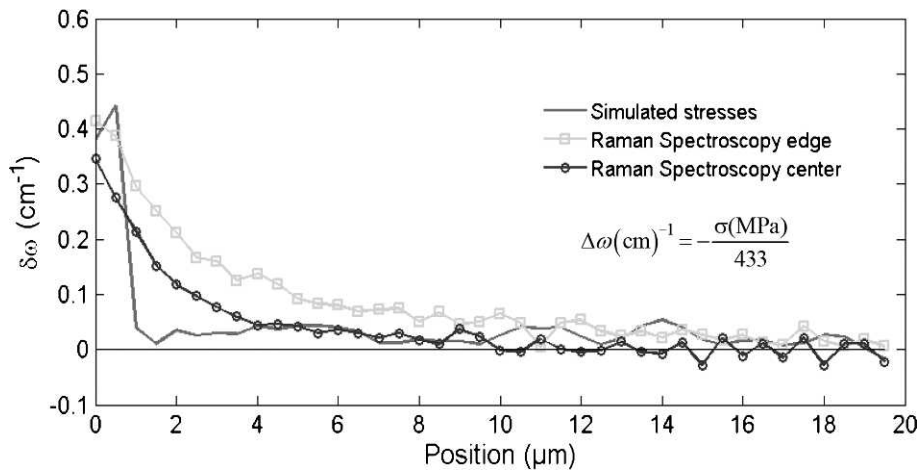


Fig. 8. Stress distribution comparisons between model predictions and Raman spectroscopy measurements.

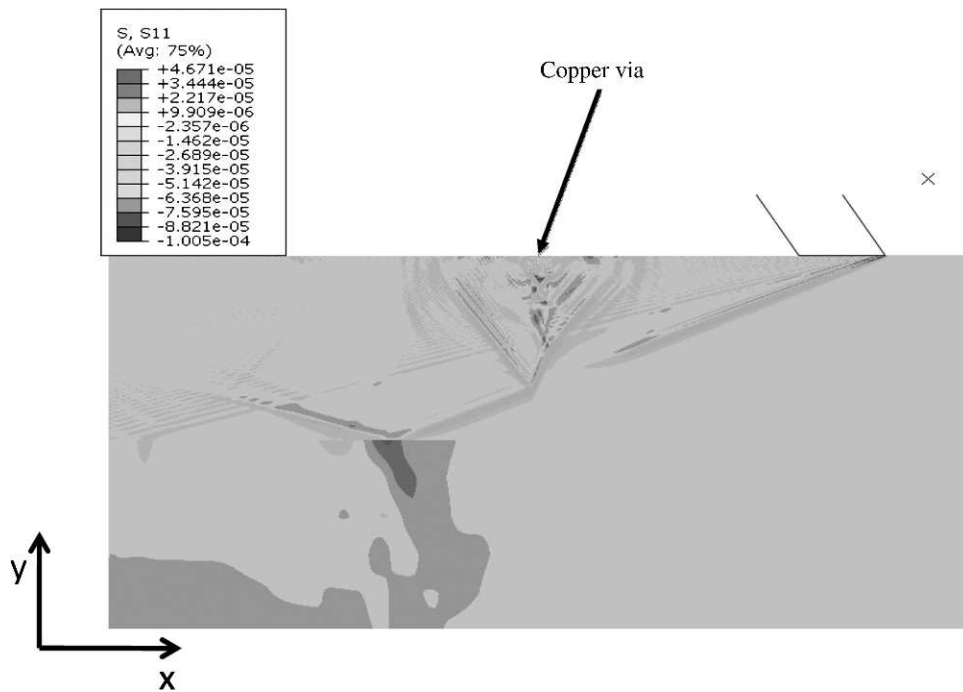


Fig. 9. Contour plot of the stress in the x direction during the cutting operation in the TSV model. Stress units are in $\text{N}/\mu\text{m}^2$.

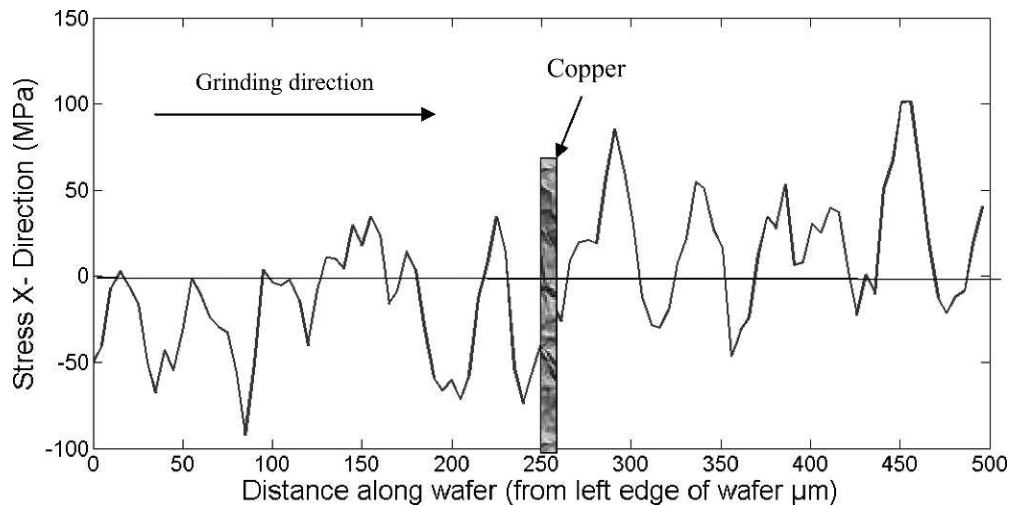


Fig. 10. Simulated stress σ_x variations at the back of the TSV wafer after grinding.

literature [15]. The amorphous layer is the superficial volume of silicon that contains random crystal orientation as a result of the intense plastic deformation caused by the grinding process. The residual stress induced in the model after grinding was measured at the back of the wafer after grinding and through the depth of the wafer. The values from the computational model were compared with the experimental data.

The residual stress at the back of the wafer after grinding reported in the literature was -150 MPa [16]. This value was originally found using Raman spectroscopy measurements for ground wafers.

Another measure for the residual stresses at the backside is the wafer warping, or bending resulting after grinding and its

release from the chuck. By measuring the postgrinding bending, one can compute the value of the residual stress using Stoney's equation [16, 17]

$$\sigma = \frac{E}{1-\nu} \frac{h^2}{6Rt} \quad (9)$$

where R is the radius of curvature of the bent wafer calculated with

$$R = \frac{a^2}{2b} \quad (10)$$

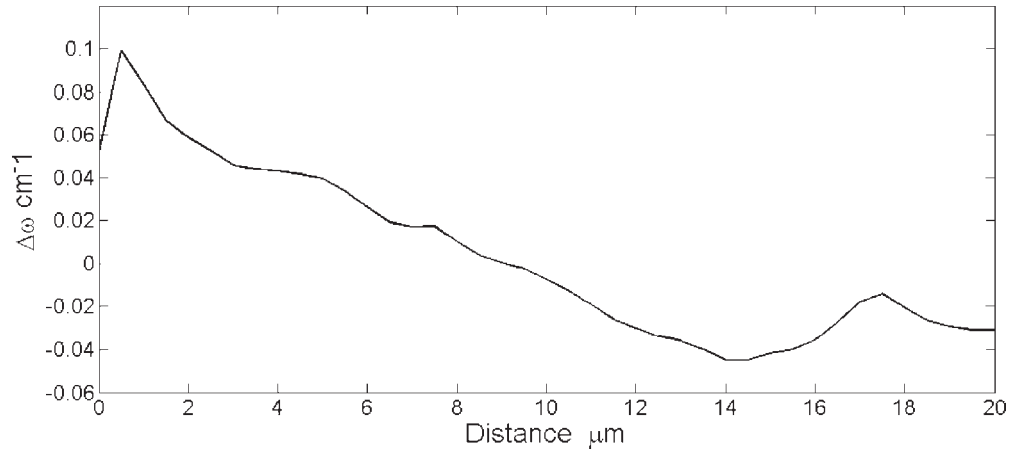


Fig. 11. Residual stresses through the TSV wafer cross section.

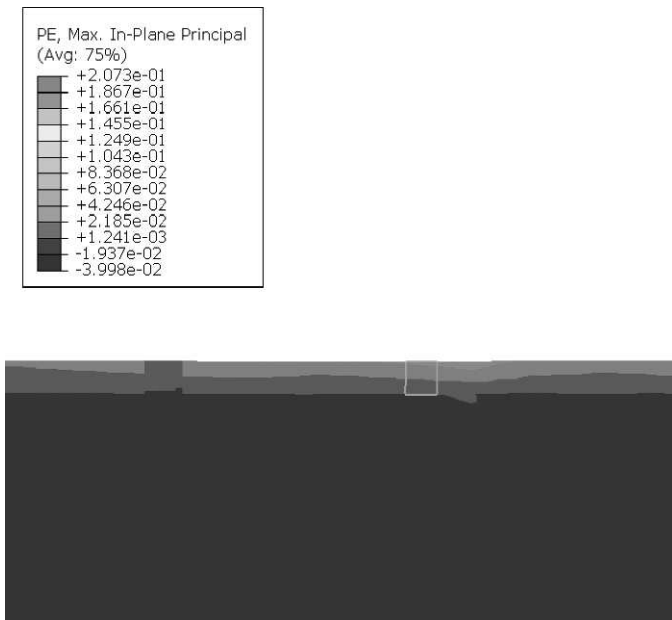


Fig. 12. Contour plot of the plastic deformation in the ground surface of the wafer. The layer thickness is approximately $0.5 \mu\text{m}$.

and $E/(1-\nu)$ is the biaxial modulus of the wafer, h is the wafer thickness, and t is the film thickness. Using Stoney's equation where $a = 90 \text{ mm}$ and $b = -160 \mu\text{m}$, the residual stress at the backside of the wafer was calculated to be -100 MPa . Therefore, the literature and experimental data led to similar results for the backside residual stresses, which ranged between -100 MPa and -150 MPa . As observed in Fig. 7, these residual stress limits match the simulation results of the residual stress distribution.

The residual stress distribution through the wafer cross section was obtained from the simulations at a depth of up to $20 \mu\text{m}$. These values were compared with the Raman spectroscopy data. The comparison showed good agreement between the simulation results and the Raman spectroscopy measurement as illustrated in Fig. 8 [16]. The graph shows good agreement between the model results and the measure-

ments at the center of the wafer for depths equal to or greater than $4 \mu\text{m}$. At depths less than $4 \mu\text{m}$, there is a small discrepancy between the model and experimental results.

B. TSV Wafer Model

A shear localization band with an angle of 30° was observed in both the silicon wafer and the copper via. Higher stresses were observed in the copper via compared with those in the silicon matrix. Fig. 9 shows the difference in the stress value in the x direction between the silicon and the copper during grinding.

The residual stress at the back of the wafer after grinding was found from the literature to range between -100 and -150 MPa . This data were originally measured using Raman spectroscopy, Stoney's calculations [16], and the output from numerical simulations of bare silicon wafers. The present simulation results for the TSV wafers were found to be in the range of -40 MPa to -90 MPa in the silicon before the TSV and in the range of -40 MPa to 100 MPa after the TSV, as shown in Fig. 10.

The simulation results for the residual stresses induced in the model through the wafer cross section at a depth of up to $20 \mu\text{m}$ were compared with Raman spectroscopy data. This data was obtained from the literature for bare silicon wafers and along the same direction in the wafer. Fig. 11 indicates a lower residual stress value compared with that from the bare silicon wafer model, which was the result of the copper via.

The extent of plastic deformation at the backside of the simulated wafer in the silicon is illustrated in Fig. 12. The figure also shows that the depth of the amorphous layer was found to be less than $5 \mu\text{m}$, which corresponds to what was reported in the literature [15]. The deformed layer is observed to have the same depth in both the copper via and the silicon wafer.

CONCLUSION

This paper presented finite element simulations of a back grinding process for silicon wafers with and without TSVs. The bare silicon wafer model was built to simulate the grinding process in the silicon, and the output results of the model were verified with experimental data from the literature. The angle of

shear band localization from the model corresponded to values reported in the literature for brittle materials. The residual stresses at the backside of the model were compared with the residual stresses reported in the literature and calculated from Stoney's equation based on wafer warping. The residual stresses through the depth of the wafer from the backside and for a depth of 20 μm were compared with the Raman spectroscopy measurements. Regarding the extent of the amorphous layer, the bare silicon wafer model showed good correlation with the experimental data. The TSV model results were compared with the experimental and simulated data for bare silicon wafers to investigate the effect of the TSV on the residual stresses and the plastic deformation. The TSV model indicated that the presence of the copper TSV leads to a reduction in the residual stresses at the backside of the wafer in the area before the TSV as well as through the wafer depth. It was also observed from the numerical simulations that the average stress in the area after the TSV becomes tensile, whereas before the TSV, the stress is compressive. The model also produced values for the thickness of the plastic deformation layer in agreement with those reported in the literature. The TSV model built on the bare silicon wafer model is able to simulate the back grinding process of the TSV wafers, and it can be used to develop a better understanding of how the parameters affect the grinding process and the quality of the ground wafers.

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REFERENCES

- [1] Z. Pei, G.R. Fisher, and J. Liu, "Grinding of silicon wafers: A review from historical perspectives," *International Journal of Machine Tools & Manufacture*, Vol. 48, No. Oct, pp. 1297-1307, 2008.
- [2] S. Chidambaram, Z.J. Pei, and S. Kassir, "Fine grinding of silicon wafers: A mathematical model for grinding marks," *International Journal of Machine Tools & Manufacture*, Vol. 43, No. 12, pp. 1595-1602, 2003.
- [3] Y. Lin and S. Lo, "A study of a finite element model for the chemical mechanical polishing process," *International Journal of Advanced Manufacturing Technology*, Vol. 23, pp. 644-650, 2004.
- [4] C.A. Chen and L. Hsu, "A process model of wafer thinning by diamond grinding," *Journal of Materials Processing Technology*, Vol. 201, No. 5, pp. 606-611, 2008.
- [5] X. Chen and W. Brian Rowe, "Analysis and simulation of the grinding process. Part II: Mechanics of grinding," *International Journal of Machine Tools & Manufacture*, Vol. 36, No. 8, pp. 883-896, 1996.
- [6] E. Ng and D.K. Aspinwall, "Modelling of hard part machining," *Journal of Materials Processing Technology*, Vol. 127, No. 9, pp. 222-229, 2002.
- [7] D. Doman, A. Warkentin, and R. Bauer, "Finite element modeling approaches in grinding," *International Journal of Machine Tools & Manufacture*, Vol. 49, No. 2, pp. 109-116, 2009.
- [8] W. Sharpe, "Mechanical properties of MEMS materials," AFRL-IF-RS-TR-2004-76, Final Technical Report, Air Force Research Labs, Rome, NY, 2004.
- [9] F. Ebrahimi and L. Kalwani, "Fracture anisotropy in silicon single crystal," *Materials Science and Engineering A*, Vol. 268, No. 8, pp. 116-126, 1999.
- [10] H. Hooputra, H. Gese, H. Dell, and H. Werner, "A comprehensive failure model for crashworthiness simulation of aluminium extrusions," *International Journal of Crashworthiness*, Vol. 9, pp. 449-463, 2004.
- [11] Dassault Systems, Abaqus, Simulia Realistic Simulation Solutions, version 6.8, www.3ds.com/products/simulia/portfolio/, 2008.
- [12] S. Kumar, G. Bae, and C. Lee, "Deposition characteristics of copper particles on roughened substrates through kinetic spraying," *Applied Surface Science*, Vol. 255, No. 1, pp. 3472-3479, 2009.
- [13] C.F.G. Gerlach, F.P.E. Dunne, D.P. Jones, P.D.A. Mills, and N. Zahlan, "Modelling the influence of filler particles on surface geometry in drawn pet film," *The Journal of Strain Analysis for Engineering Design*, Vol. 31, No. 1, pp. 65-73, 1996.
- [14] A.G. Atkins, "Modelling metal cutting using modern ductile fracture mechanics: Quantitative explanations for some longstanding problems," *International Journal of Mechanical Sciences*, Vol. 45, No. 2, pp. 373-396, 2003.
- [15] E. Brinksmeier, Y. Mutlugünes, F. Klocke, J. Aurich, P. Shore, and H. Ohmori, "Ultra-precision grinding," *CIRP Annals, Manufacturing Technology*, Vol. 59, pp. 652-671, 2010.
- [16] J. Chen and I.D. Wolf, "Study of damage and stress induced by backgrinding in Si wafers," *Semiconductor Science and Technology*, Vol. 18, pp. 261-268, 2003.
- [17] G.C.A.M. Janssen, M.M. Abdalla, F. van Keulen, B.R. Pujada, and B. van Venrooy, "Celebrating the 100th anniversary of the Stoney equation for film stress: Developments from polycrystalline steel strips to single crystal silicon wafers," *Thin Solid Films*, Vol. 517, No. 3, pp. 1858-1867, 2009.